HEIQ XMEMBRANE PROJECT TEAM

Current result and decision report

Integrated system, membrane, material and coupon-cell evidence

31 August 2026 · Version 8

Decision basis for the project team

This report consolidates the current model closure, the post-ERV full-duty and percentage studies, common-voltage results, system and figure-of-merit ledgers, membrane/material screens, and the centimetre-scale vapour-cell design. It supersedes earlier team-facing interpretations for current decisions, while retaining their machine-readable evidence as provenance.

Current milestone. The broad modelling phase has reached its stop criterion. The next phase is material and subsystem feasibility: acquire a same-sample membrane-property bundle, build the controlled-flow coupon cell, and close the receiver/thermal/hardware ledger. No material, module, technology or net full-system saving has been validated.

Primary equal-service reference	Passive ERV plus remaining AC: 543.741 kW at the declared O80 outdoor-air duty
Leading first experiment	Post-ERV EO toward O30 in a controlled-flow 25-cm^2 coupon cell
Architecture status	O30 is sampled below 2 V on one common bus; full remaining post-ERV duty is not demonstrated on one common bus
Release status	Model-phase delivery is internally checkable; immutable release snapshot remains open because the working tree is not frozen
Claim status	Conditional model and experiment-planning evidence only

Abstract

The present evidence supports one clear phase decision: stop broad parameter expansion and begin controlled material and subsystem testing. At the 150,000-cfm O80 duty, the equal-service electrical reference is 543.741 kW for passive ERV plus the remaining AC load, compared with 1,166.4 kW for AC alone. A post-ERV, common-voltage O30 coordinate gives a known subtotal of 381.327 kW and therefore a conditional 162.414-kW window relative to passive ERV+AC. This is not a realised saving because receiver/sink energy, EO heat rejection, electrodes, manifolds, leakage, durability and measured material performance remain open. O50 requires no post-ERV EO duty on the declared absolute-humidity basis; O30 closes 72.700% of the remaining post-ERV moisture gap and is the strongest first experimental target.

The full-duty and percentage studies show that the historical 10% coordinate actually closes 22.193% of the remaining post-ERV duty. Uniform- current sensitivity rows can close full duty below 2 V, but the independent one-common-bus replay has no confirmed full-duty point on the sampled 0.5–2.0-V grid; 21 refined rows are physical no-solutions and 28 remain numerical/local unresolved. This is not a global no-go proof. At O30, sampled common-bus feasibility survives from $D = 10^{-14}$ to 10^{-9} m^2/s, but adverse diffusion erodes the energy window and reverses it at the highest sampled diffusivity.

The measured-data requirement is a coupled, same-sample bundle: sorption and hysteresis, chemical water mobility/permeance, electro-osmotic drag, wet conductivity and compressed ASR, thickness/swelling/mechanics, coupled net water flux, and receiver/electrode compatibility. Nafion is a useful numerical reference but does not match the required drag-property vector and is not a validated candidate. The proposed first fixture uses a 50×50-mm active aperture, an approximately 80×80-mm coupon, 110×110-mm outer plates, interchangeable 1/2/4-mm gaps and controlled flow. Polarity reversal, two-sided water balance, pressure control, blanks and gap/flow repeats are required to separate EO transport from static humidity/temperature mismatch, films, edge leakage and wall inventory.

162.4 kW
Conditional O30 window
Known common-voltage subtotal before open losses

72.70%
Remaining duty at O30
O50 requires no post-ERV EO duty

25 cm^2
First active aperture
Controlled flow, ABBA and two-sided balance

Contents

Abstract **2**

1 Executive conclusion and decision map **4**
1.1 Current integrity state . . . 4

2 System architecture and equal-service comparison **4**

3 O50, O30 and the meaning of percentage **5**
3.1 O50 . . . 5
3.2 O30 . . . 6
3.3 The historical 10% point . . . 6

4 Increasing EO share after the ERV **6**
4.1 Full-duty uniform-current sensitivity . . . 7

5 Common-voltage findings **8**
5.1 Diffusivity sensitivity . . . 8

6 Figure of merit and energy budget **10**

7 Membrane and material findings **10**
7.1 Direct EO and post-ERV parameter regions . . . 11
7.2 Nafion interpretation . . . 11
7.3 Same-sample data required before ranking . . . 12

8 EO vapour coupon-cell dimensions **12**
8.1 Signal and flow modes . . . 12

9 Boundary sensitivity and error control **13**

10 Open gates **14**

11 Recommended next experiments **14**
11.1 Experiment 1: fixture qualification and blank closure . . . 15
11.2 Experiment 2: EO identification with matched zero-current control . . . 15
11.3 Experiment 3: same-sample property bundle . . . 15
11.4 Experiment 4: receiver and thermal ledger . . . 15
11.5 Experiment 5: measured-data topology replay . . . 15

12 Provenance and claim boundary **15**
12.1 Permitted conclusions . . . 16
12.2 Prohibited conclusions . . . 16

A Compact numerical register **16**

1 Executive conclusion and decision map

WHAT IS DECIDED NOW

GO: material and coupon work. Begin a same-sample material search and the controlled-flow vapour-cell experiment. Use O30 after the passive ERV as the first system-relevant transport target. Continue modelling only to replay measured candidate data, pre-register experiments, or close a named subsystem.

OPEN: architecture and receiver. Do not yet choose one-bus full duty, voltage zoning, staged EO, direct EO inside the ERV, or a liquid/solid sink. Those decisions require measured film behaviour, receiver work, heat rejection, pressure drop, leakage, electrodes and durability.

NO-GO: validation language. Do not state that 162.4 kW or any larger model window has been saved, that full duty is impossible, that Nafion or an inverse-design vector is a candidate membrane, or that the technology is validated. The working tree must also be frozen before an immutable release claim is made.

Decision or question	Status	Current basis
Equal-service energy comparator	CLOSED	Use 543.741 kW passive ERV+AC as the primary reference
O30 common-voltage mechanism below 2 V	SUPPORTED	Confirmed sampled points, including 1.00 V, with local current sharing
Full post-ERV duty on one bus below 2 V	OPEN	No confirmed sampled point; physical and unresolved rows coexist
Broad model phase	STOP	Core balances, signs, numerical replays and decision coordinates are sufficient for handoff
Material candidate ranking	NOT READY	Zero of nine required same-sample data rows is complete
Receiver and thermal contract	OPEN	Receiver/sink, disposal/regeneration and EO heat rejection are not closed
Practical module limits	OPEN	Team must set area, pressure-drop, footprint, cost and lifetime limits
Net system or product claim	NOT ALLOWED	No measured full ledger or validated material/module
Material search and coupon cell	GO	Use the joint-property schema and controlled-flow boundary protocol

1.1 Current integrity state

The current final-delivery checker classifies the model-phase package as deliverable and the material search as ready to start. It keeps the immutable release snapshot open because the repository is dirty, and it keeps candidate ranking, full-duty single-bus maturity, receiver closure and full-system claims open. Audit PASS means declared balances, identities, hashes and schema checks passed at that snapshot; it is not a technology pass.

2 System architecture and equal-service comparison

ERV PRECEDES EO IN THE LEADING POST-ERV ARCHITECTURE

The leading near-term chain is

$$\text{outdoor air} \longrightarrow \text{passive ERV} \longrightarrow \text{EO polishing} \longrightarrow \text{remaining AC sensible/latent service.}$$

EO therefore acts on the finite moisture duty that remains after the ERV. This is distinct from direct EO integrated into the ERV core, and from an EO-first architecture without a passive latent exchanger.

System	AC [kW]	Pre-dryer [kW]	Total [kW]	Evidence boundary
AC only	1,166.400	0	1,166.400	Calibrated reference anchor
Passive ERV + AC	509.381	34.359	543.741	Modelled passive equal-service reference
Passive ERV + post-ERV EO to O30 + AC	316.910	64.417	381.327	**Conditional incomplete ledger**; receiver/sink and several hardware terms omitted
Direct active EO in ERV, full humidity duty + AC	44.438	63.419	107.856	**Conditional architecture screen**; unmeasured inverse-design vector and incomplete coupling

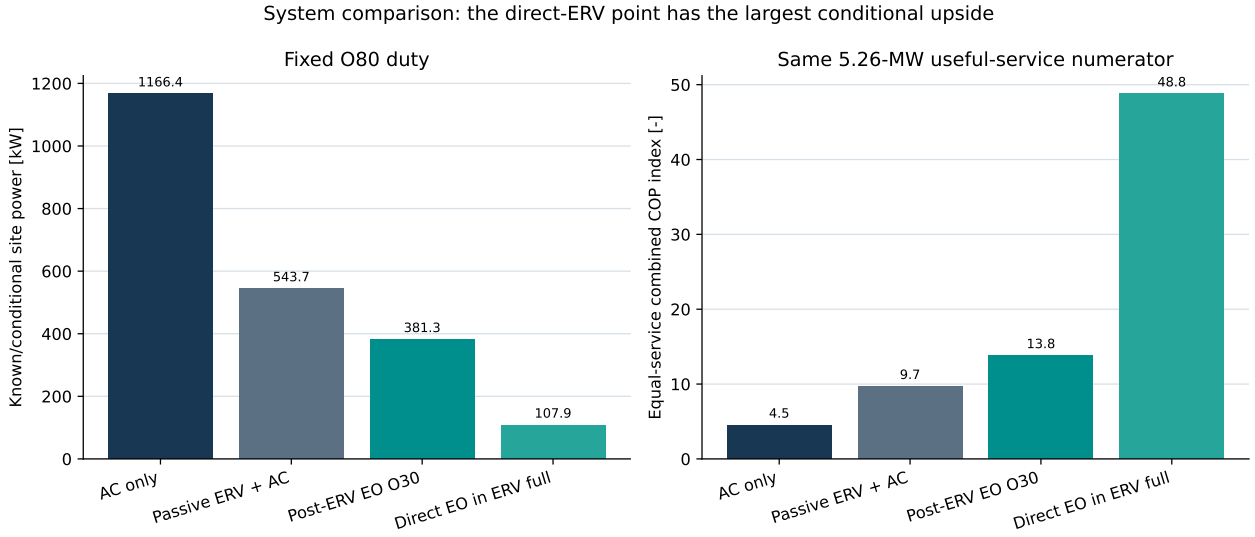


Figure 1: Current combined system comparison on one equal outdoor-to-supply water duty. The AC-only and passive-ERV rows are references. Both EO rows are conditional ledgers: they include the declared known electrical terms but exclude open receiver/sink, thermal and hardware penalties.

Relative to passive ERV+AC, the current common-voltage O30 subtotal is lower by 162.414 kW, or 29.87%. Relative to AC alone it is lower by 785.073 kW. These differences are *available conditional budgets*, not measured savings. Every omitted positive load consumes the budget:

$$P_{\mathrm{EO,all-in}} = P_{\mathrm{grid}} + P_{\mathrm{module\,fan}} + P_{\mathrm{receiver/sink}} + P_{\mathrm{heat\,rejection}} + P_{\mathrm{BOP}}.$$

Why EO follows ERV here. The passive ERV removes 3,137.65 kg/h from the outdoor-to-supply water duty before EO is invoked. This reduces the EO duty and makes O50 a zero-EO case. The direct active-ERV architecture remains a model-level research alternative, but it has a different boundary, different membrane preference and less subsystem closure.

3 O50, O30 and the meaning of percentage

USE FINITE REMAINING DUTY, NOT AMBIGUOUS RELATIVE HUMIDITY LABELS

3.1 O50

At the frozen O80 outdoor state and passive-ERV reference, the ERV outlet is already drier than the O50 absolute-humidity coordinate. O50 therefore requires **no post-ERV EO duty**. The O50 bar is not evidence for an EO saving; it is the passive-ERV reference state.

3.2 O30

O30 removes 1,228.27 kg/h after the passive ERV and closes 72.7003% of the finite moisture gap between the ERV outlet and the final supply humidity ratio. It is the clearest first target because it:

- creates a large but finite and interpretable water duty;
- has confirmed sampled one-common-bus solutions below 2 V;
- retains a sizeable conditional receiver/sink budget;
- exposes the decisive diffusivity, film and material interactions without requiring the full-duty pinch to close.

3.3 The historical 10% point

The previously named 10% reduction removes 374.94 kg/h. Relative to the finite post-ERV-to-supply duty, it is **22.1925% completion**. Both coordinates are retained for traceability; the historical result must not be silently relabelled.

Coordinate	EO water [kg/h]	Remaining-duty completion [%]	Fixed-trajectory area [m^2]	Maximum voltage [V]
Historical 10%	374.94	22.1925	1,858.5	0.503
O30	1,228.27	72.7003	6,089.1	1.314
Full remaining duty	1,689.50	100.0000	8,376.2	1.895

The area/voltage values above are the fixed Terminal-1 uniform-current sizing trajectory. They do not establish a common-bus module. The independent common-voltage results are reported separately in Section 5.

4 Increasing EO share after the ERV

THE ENERGY TREND IS CONDITIONAL AND THE ENDPOINT CHANGES CONTROL REGIME

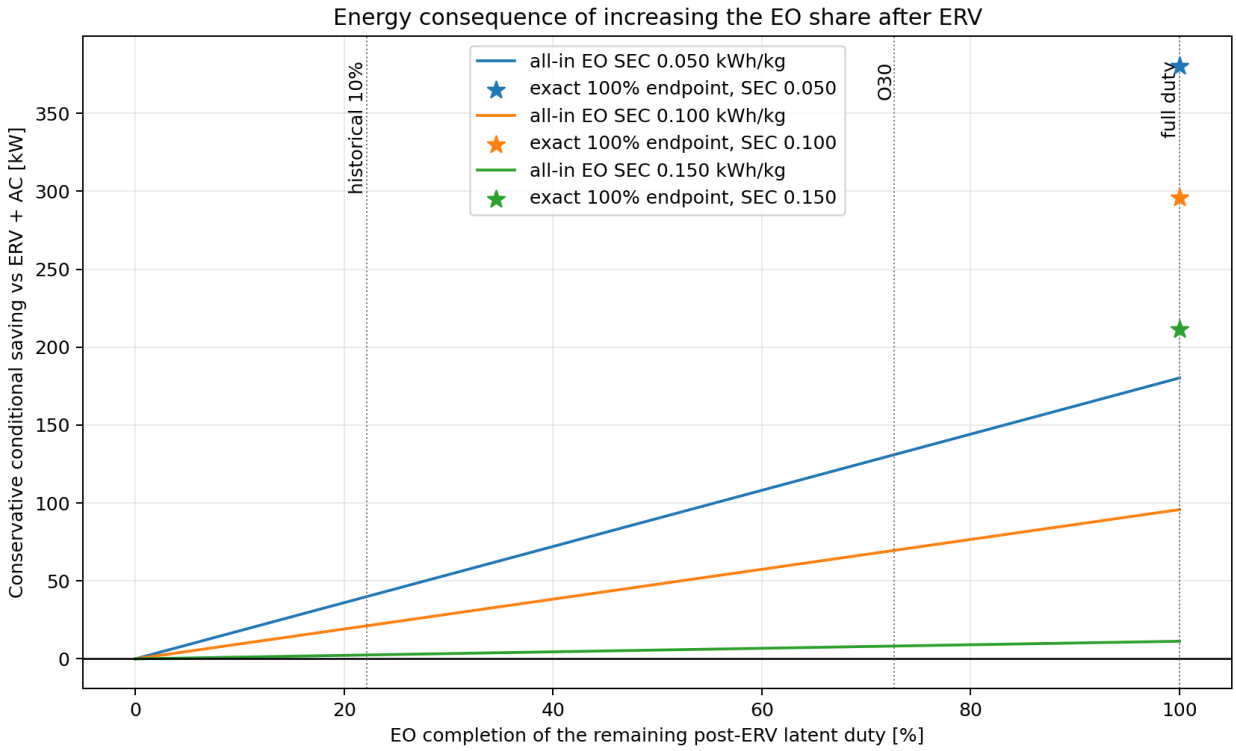


Figure 2: Conditional energy effect as EO closes an increasing fraction of the remaining post-ERV duty. The exact 100% endpoint is a different coil-control regime and must not be connected continuously to the one-sided near-endpoint curve.

Below the exact full-duty endpoint, the screening coil remains on its saturated-coil branch. With zero EO process-heat leakage, the conservative energy window grows approximately linearly with treated water when all-in EO SEC is below 0.1567 kWh/kg, is neutral near that value, and becomes an increasing loss above it.

All-in EO SEC	Historical 10%	O30	Full duty
	Conservative conditional window [kW]		
0.050 kWh/kg	40.01	131.06	180.27
0.100 kWh/kg	21.26	69.64	95.80
0.150 kWh/kg	2.51	8.23	11.32

The exact endpoint has a calculated 0.2752-kWh/kg break-even because the remaining coil becomes sensible-only. Infinitesimally above the endpoint, the screening logic returns to the latent-control branch and the conservative budget is 0.1567 kWh/kg. Use the lower one-sided budget for decisions until a real coil/control sequence validates the ideal endpoint transition.

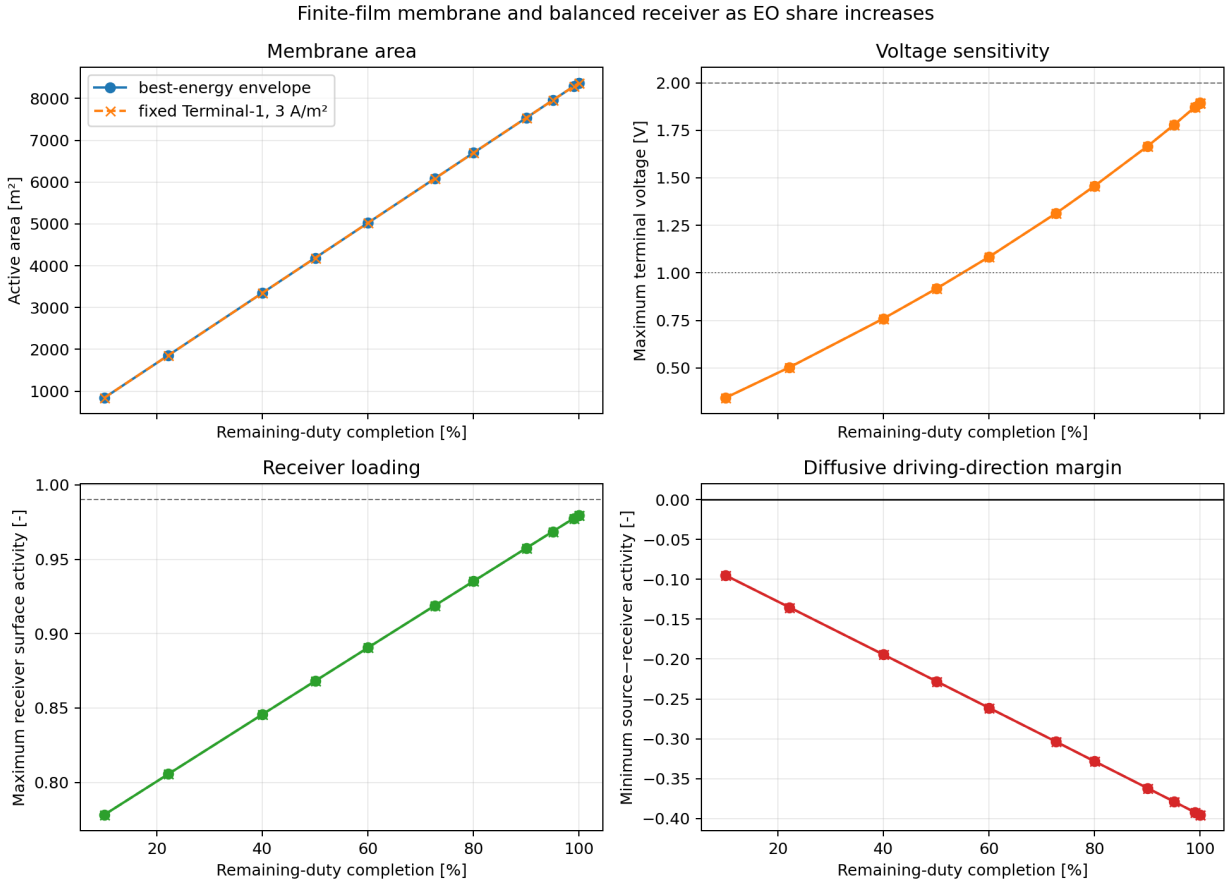


Figure 3: Fixed-coordinate membrane area, voltage and receiver activity versus remaining-duty completion. This is a re-sized trajectory at each target, not one fixed installed module.

4.1 Full-duty uniform-current sensitivity

The best target-reaching balanced-receiver row in the uniform-current study uses 8,376.2 m^2, 3.0 A/m^2 and 1.895 V. It leaves 216.80 kW on the conservative near-endpoint branch before receiver/sink work, or 417.00 kW on the ideal exact-endpoint branch. The receiver/sink SEC room is 0.1283 kWh/kg. This is useful as a thermodynamic and sizing sensitivity, not evidence that one electrical bus or a practical receiver can realise the point.

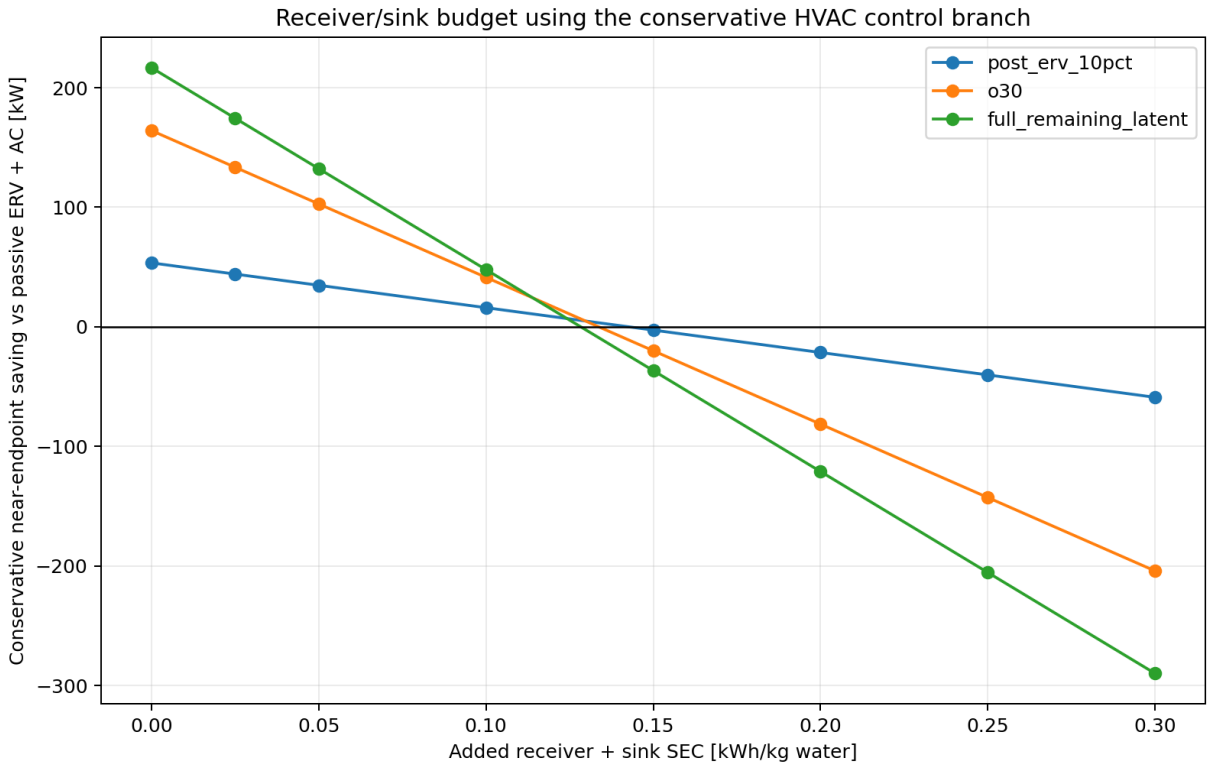


Figure 4: Conditional receiver/sink SEC budget for full post-ERV duty. Every receiver, regeneration, disposal, pumping and heat-rejection term must fit inside the remaining budget for the energy sign to survive.

5 Common-voltage findings

ONE BUS WITH LOCAL CURRENT SHARING IS STRICTER THAN A UNIFORM-CURRENT SWEEP

The common-voltage topology applies one terminal voltage while allowing local parallel current sharing. No current-density acceptance ceiling and no area acceptance cap are used. Feasibility requires endpoint, water, voltage and thermodynamic closure.

Replay	Completion [%]	Voltage [V]	Area [m^2]	j range [A/m^2]	Status
50%	50.000	1.000	2,468.35	3.646–6.879	Converged target
60%	60.000	1.000	3,708.38	2.432–6.322	Converged target
67.5%	67.500	1.000	5,269.67	1.520–5.905	Converged target
O30	72.700	1.000	7,191.41	0.887–5.616	Converged target
O30 balanced-anchor voltage	72.700	1.314	3,461.40	3.044–8.427	Converged target
Full duty	100.000	1.000	–	0.064–10.964	No physical common-voltage solution at this coordinate
Full duty	100.000	2.000	–	2.358–10.006	Numerical/local failure; not a no-go proof

5.1 Diffusivity sensitivity

The refined one-factor study varies D while freezing the remaining inverse-design properties. O30 has confirmed sampled voltages at 1.00–1.25 V for all seven D coordinates. At $D = 10^{-14}$ m^2/s, the selected 1.00-V row uses approximately 7,174 m^2, consumes 30.06 kW known EO+fan power and leaves a conditional 162.42-kW window. At $D = 10^{-9}$ m^2/s, the selected known power rises to 373.89 kW and the conditional window becomes -181.41 kW.

For full remaining duty, none of 49 refined sampled voltages is confirmed: 21 are physical no-solutions and 28 are numerical/local unresolved. Across both segment counts, 98 flagged rows were retried at a 100-times higher configured numerical bound; none recovered full closure. The allowed wording is therefore: *no confirmed feasible sampled common-voltage point*

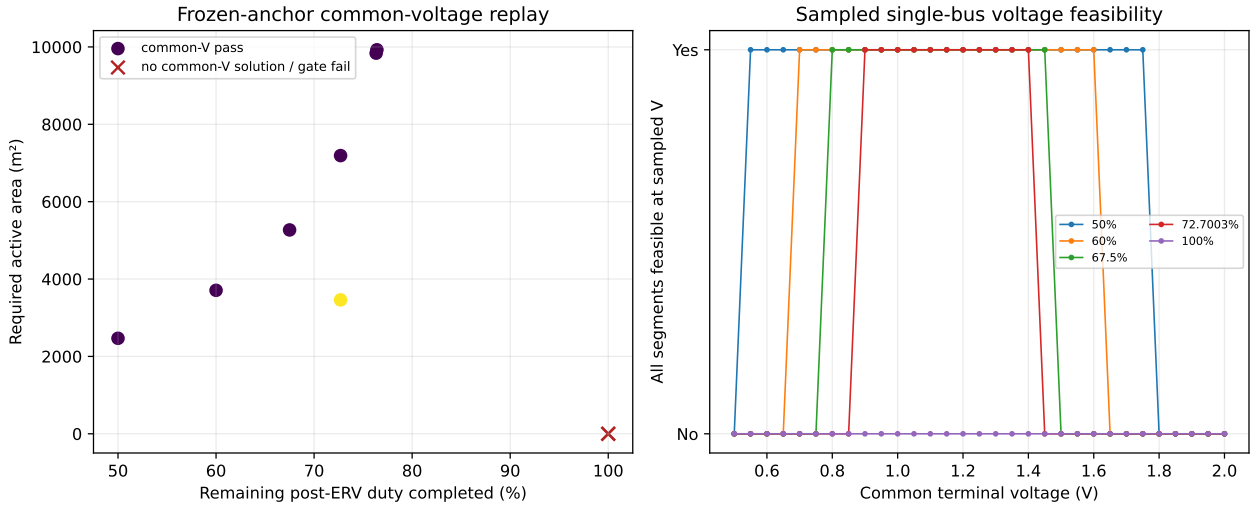


Figure 5: Project-closure replay of the common-voltage window. O30 and lower completion targets have confirmed sampled points. Full duty has no confirmed sampled point in the frozen range and remains an open architecture question.

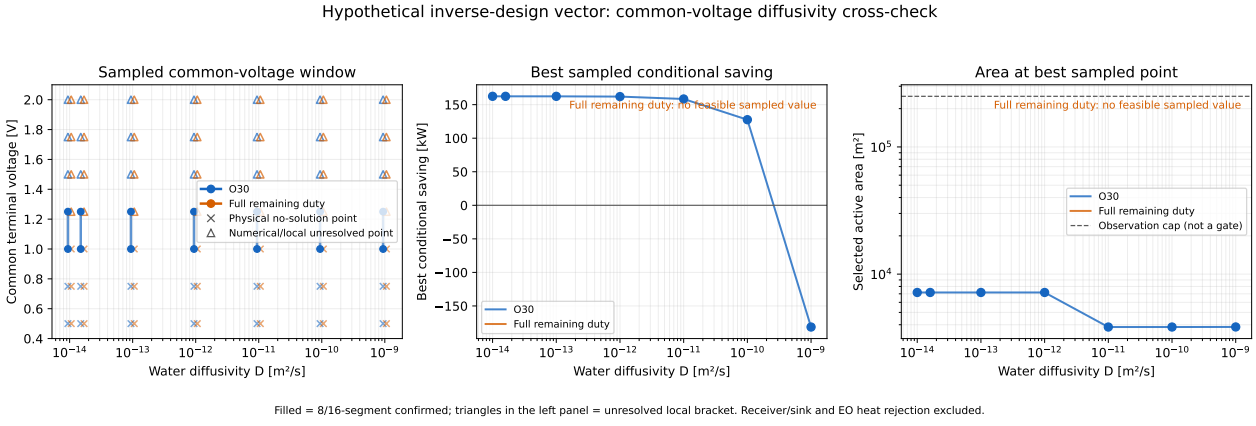


Figure 6: Common-voltage diffusivity cross-check. Crosses denote sampled physical no-solutions and open triangles unresolved numerical/local high-bracket rows. No sampled value is not a global proof of impossibility.

was found below 2 V at the frozen coordinate. Voltage zoning, staging or changed receiver/path physics belongs after measured material data.

6 Figure of merit and energy budget

FOM IS THE RECIPROCAL OF SEC ONLY AFTER THE POWER BOUNDARY IS NAMED

$$\mathrm{FOM} = \frac{\dot{m}_{\mathrm{water}}}{P_{\mathrm{device}}} \quad [\mathrm{kg/kWh}], \qquad \mathrm{SEC} = \frac{P_{\mathrm{device}}}{\dot{m}_{\mathrm{water}}} \quad [\mathrm{kWh/kg}], \qquad \mathrm{FOM} = \mathrm{SEC}^{-1}.$$

Device coordinate	Water [kg/h]	Known power [kW]	Known SEC [kWh/kg]	Known FOM [kg/kWh]	Boundary
Post-ERV EO to O30, 1.00 V	1,228.27	30.057	0.02447	40.864	EO grid plus idealised module fan
Direct EO in ERV, full duty, 0.25 V	4,827.15	63.419	0.01314	76.116	Reclassified active core plus idealised fan

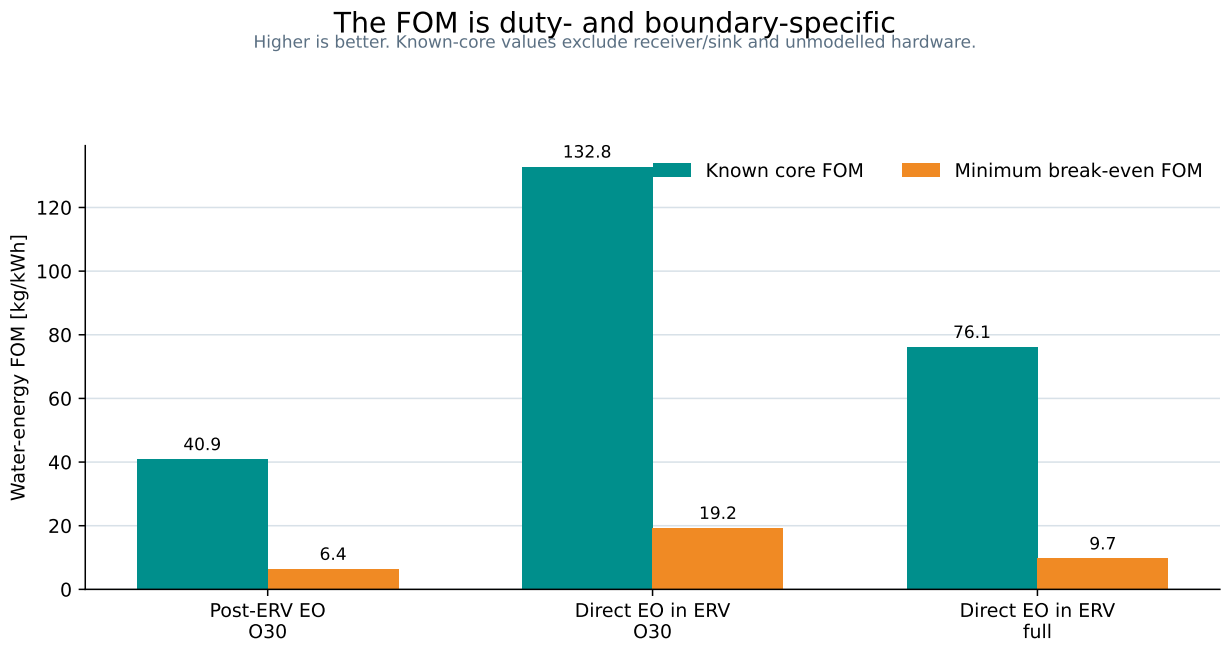


Figure 7: Known-core FOM and the minimum all-in FOM needed to retain a positive conditional margin. The open gap between known and break-even performance is an allowance for missing loads, not evidence that those loads are small.

For the post-ERV O30 common-voltage point, the minimum break-even all-in FOM is 6.382 kg/kWh and the unallocated SEC budget is 0.13223 kWh/kg. The current known-core FOM is attractive only if the receiver, thermal and balance-of-plant terms fit inside that allowance.

Using the complete outdoor-to-supply water duty as a common numerator, the system water-energy FOMs are 4.139 kg/kWh for AC only, 8.878 for passive ERV+AC, 12.659 for post-ERV EO O30, and 44.755 for the direct active-ERV full-duty screen. The EO system values remain conditional incomplete ledgers. Twelve of thirteen requested FOMs can be calculated or parameterised; payback cannot be stated without CAPEX and incremental maintenance.

7 Membrane and material findings

THE TARGET IS A COUPLED PROPERTY VECTOR, NOT ONE FAVOURABLE SCALAR

7.1 Direct EO and post-ERV parameter regions

The direct-EO 2-V screen has 178 membrane-gate passes among 2,508 physical coordinates; 108 remain at the preferred 1.8-V boundary. Those are discrete sampled supports, not continuous tolerance boxes. With a 0.85-V combined fixed loss, none is positive against the modern comparator before receiver energy. A deterministic 0.20-V fixed-loss scenario has 15 positive rows, still before receiver and balance-of-plant terms.

For EO after the ERV, 646 of 1,122 dense D–j rows are conditionally energy-positive on the declared coefficient basis. This does not imply that 646 materials exist: all other properties were frozen.

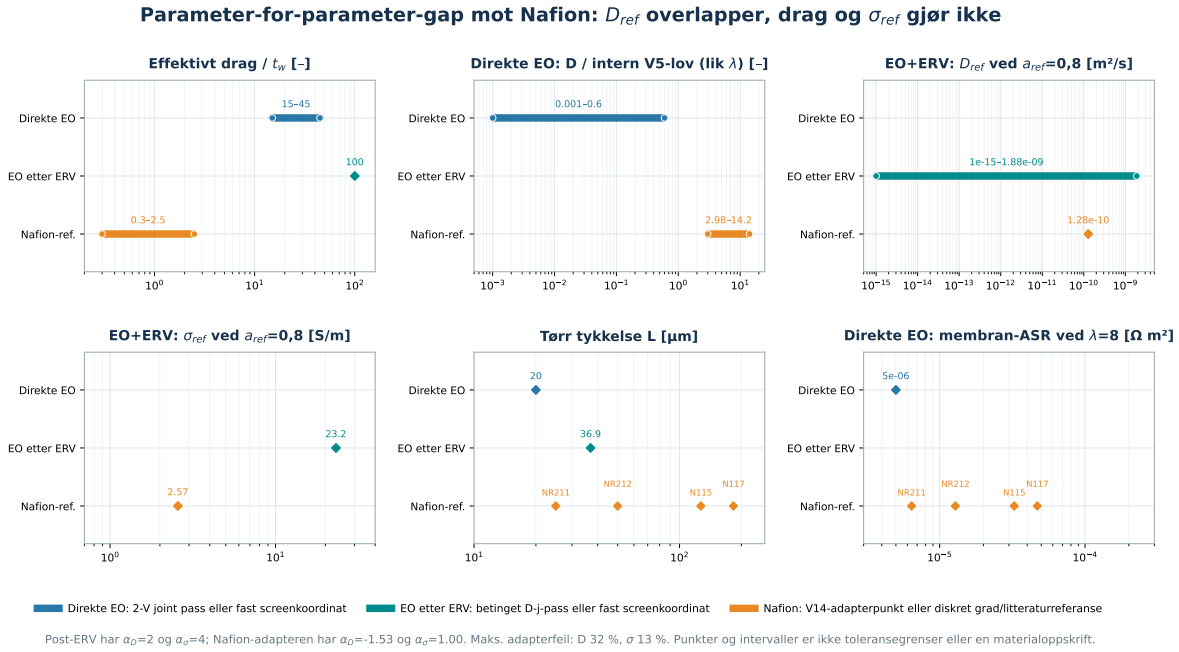


Figure 8: Simultaneous property gap relative to the declared Nafion reference. Marginal axis overlap must not be combined into a fictitious joint material vector.

7.2 Nafion interpretation

Nafion's declared drag envelope is approximately 0.3–2.5 mol H_2O/mol charge, far below the direct-EO drag region in the current screen. Its literature diffusivity can lie on a sampled post-ERV D axis, but its conductivity magnitude and activity exponents differ from the frozen inverse-design model. The adapter has non-negligible fit errors and some active-ERV local states lie outside its declared hydration domain.

A matched zero-current control is especially instructive. At the selected Nafion-reference coordinate, the active row attributes 91.63% of net water transfer to diffusion and consumes 0.444 kW for EO. The exact passive control requires only 2.653% more area and uses 0.493 kW less electricity. Consequently, that apparent benefit is primarily passive membrane transfer, not an EO energy result.

Material non-gates. Nafion is not the target. There is no current density acceptance gate at 21 A/m^2, area is an observation rather than a pass/fail cap in this phase, and a drag number near 100 is not sufficient by itself. Do not mix properties from different samples, conditioning states or literature schemas.

7.3 Same-sample data required before ranking

ID	Property or measurement	Decision role
MAT-01	Sorption isotherm, hysteresis and water capacity	Converts RH into chemical loading/work and defines the valid domain
MAT-02	$D(a, T)$ or chemical mobility/permeance with definition	Separates kinetics from capacity and quantifies passive co-/back-diffusion
MAT-03	EO drag $n_d(a, T, j)$	Sets polarity-odd water transported per charge
MAT-04	Wet through-plane conductivity	Sets bulk ohmic loss
MAT-05	Compressed membrane plus contact ASR	Prevents optimism from bulk conductivity alone
MAT-06	Dry/wet thickness, swelling, creep and mechanics	Couples permeance, resistance, gap control and lifetime
MAT-07	Coupled two-sided net water flux	Tests whether the property bundle coexists under model-relevant gradients
MAT-08	Receiver/electrode wetting, corrosion, flooding and drainage	Closes interface compatibility and regeneration needs
MAT-09	Sample, batch, ion form, conditioning and raw-data provenance	Makes the candidate auditable and procurable

Post-ERV polishing and direct-ERV integration can prefer different materials. Post-ERV EO benefits from suppressing adverse chemical permeance. Direct ERV can exploit favourable passive transfer before a pinch, so minimising permeance indiscriminately can remove useful free transport.

8 EO vapour coupon-cell dimensions

CENTIMETRE-SCALE, CONTROLLED-FLOW AND RECONFIGURABLE

Element	Starting value	Reason
Maximum active aperture	50×50 mm = 25 cm^2	Maximises signal while retaining a manageable fixture
Additional square masks	30×30 and 20×20 mm	Area scaling and lower-signal checks
Identifiability inserts	40×40 mm and equal-area 8×50 mm	Separates area/perimeter and orientation effects
Membrane coupon	Approximately 80×80 mm	15-mm land per side for seals, guard and handling
Outer fixture	Approximately 110×110 mm	Additional land for bolts and ports
Symmetric gaps	1, 2 and 4 mm; begin at 2 mm	Direct film and boundary sensitivity
Flow frame	8–10 mm insulating frame	Starting structural and port depth
Backing/load plate	12 mm metal; 16 mm if load-bearing PEEK	Conservative deflection screen pending measured torque/flatness
Nominal high-flow mode	6 actual L/min per side at 2-mm gap	Approximately 1 m/s, Re about 236, ideal straight-channel drop about 2.9 Pa
Swept volume	5.00 mL active; 6.22 mL with headers/ports	Keep added dead volume at or below about 1.3 mL per side

The plate screen limits bow to 20 μm, or 2% of the 1-mm gap, across a 50-mm unsupported square. Gas pressure alone is not governing. Under a deliberately conservative 1-MPa equivalent uniform fixture-load surrogate, calculated minimum thicknesses are 15.4 mm PEEK, 5.8 mm aluminium and 4.2 mm 316L. These are screening values, not a gasket-contact calculation; the assembled cell needs a torque, loaded-gap and flatness map.

8.1 Signal and flow modes

The 25-cm^2 gross EO signal screen spans approximately 5.04–840.2 mg/h. With 0.1 RH-percentage-point expanded uncertainty for each humidity channel and a signal-to-uncertainty gate of five, low-drag signals would require flows below the 6-L/min film-characterisation condition. Use separate modes:

- **high-flow film mode:** nominally 6 L/min per side, repeated at 3 and 12 L/min and at 1/2/4-mm gaps;

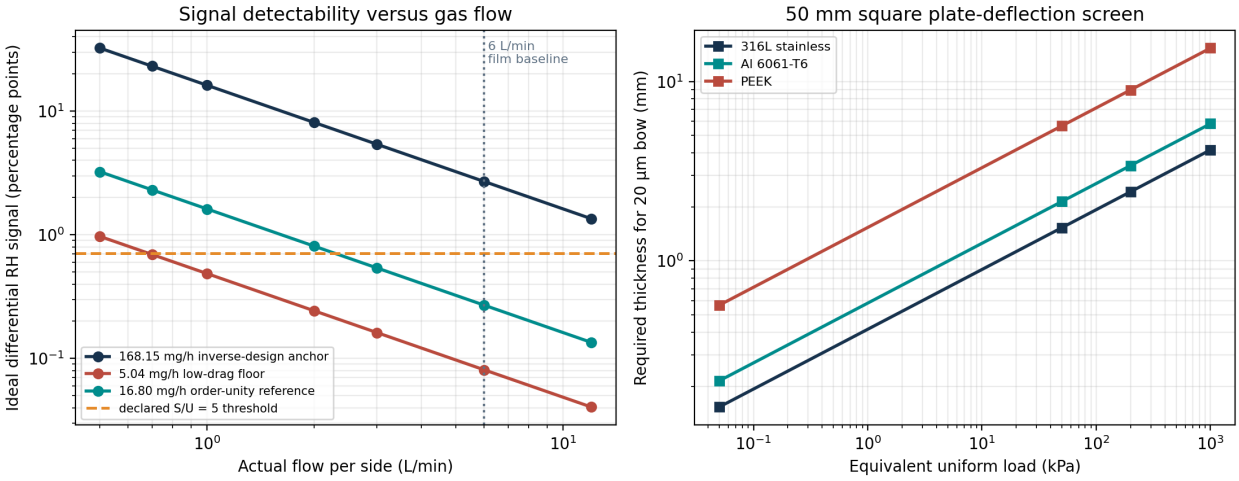


Figure 9: Coupon-cell dimension trade-offs. The 25-cm^2, 2-mm-gap fixture is the nominal starting point; interchangeable masks and gaps are part of the measurement design, not optional refinements.

- **low-flow/integrating signal mode:** approximately 0.69 L/min for the 5.04-mg/h floor or 2.29 L/min for the 16.80-mg/h order-unity reference, unless a more sensitive integrating water measurement is used.

A shallow closed chamber cannot hold fixed RH under transport. In the gas-only 5-mL chamber, the low-drag 5.04-mg/h reference reaches a 5-Pa differential in 0.063 s. Even a 5-L reservoir extends that ideal time only to 63.4 s. Closed operation is therefore a transient inventory experiment with pressure compliance/control, not a steady boundary.

9 Boundary sensitivity and error control

POLARITY REVERSAL REMOVES STATIC BIAS, NOT FILMS OR DYNAMIC DRIFT

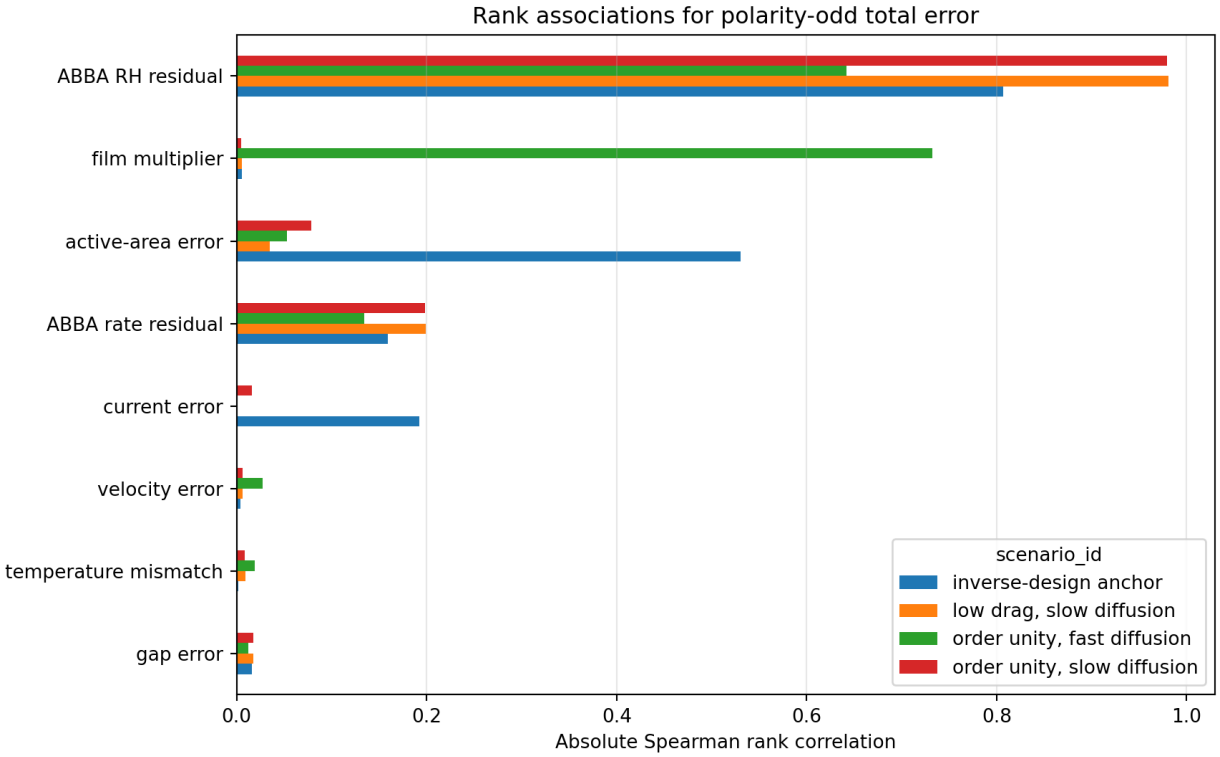


Figure 10: Ranked boundary and nuisance sensitivities in the declared stress space. The ranges are experiment-design scenarios, not calibrated probability distributions or confidence limits.

At the nominal 25-cm^2, 2-mm, 32°C condition:

- a static +0.1 RH-percentage-point inlet mismatch changes the fast-diffusion one-way result by about +10.2%, while the ideal polarity-odd estimator nearly rejects it;
- a +0.1-K side-to-side temperature mismatch changes the same one-way result by about -28.9%, while a stable reversal again nearly rejects the static part;
- changing the gap from 2 to 1 mm shifts the fast-diffusion odd signal by about -2.0%; changing to 4 mm shifts it by about -10.1%;
- high-diffusion cases are dominated by the unresolved film multiplier, while slow-diffusion cases are dominated by humidity drift between polarities.

ABBA ordering rejects static and linear drift only when the boundary is stable over the block. Polarity-correlated humidity, wall inventory, nonlinear drift, hysteresis and settling can survive the odd estimator and mimic EO. Therefore:

1. feed both sides from a split common inlet for equal-RH work;
2. cross-switch the humidity reference and randomise or mirror ABBA block order;
3. record source loss and receiver gain independently and require water closure within combined uncertainty;
4. control membrane differential pressure to the declared ±5-Pa target and deliberately challenge it separately;
5. run impermeable-foil blanks before and after each block;
6. report one-way, polarity-even and polarity-odd rates separately;
7. repeat flow and gap, and fit area plus perimeter using at least five inserts including an equal-area/different-perimeter pair.

The original three square masks alone give zero residual degrees of freedom for an area-plus-perimeter fit and an area/perimeter correlation of 0.995. The recommended five-insert set yields two residual degrees of freedom and reduces, but does not eliminate, the correlation.

10 Open gates

WHAT STILL PREVENTS A DESIGN OR ENERGY CLAIM

Gate	State	Required closure evidence
Candidate data completeness	OPEN	One same-sample MAT-01–MAT-09 dataset with raw points and provenance
Receiver boundary and SEC	OPEN	Chosen gas/liquid/desiccant sink; measured transfer, pumping, regeneration or disposal and compatibility
EO heat rejection	OPEN	Conjugate energy balance with transported-water enthalpy assigned once
Full-duty common bus	OPEN	Measured candidate replay and, if warranted, staged/zoned topology
Films and interfaces	OPEN	Flow/gap/temperature-resolved source and receiver exchange coefficients
Hardware/BOP	OPEN	Electrodes, contacts, converter topology, manifolds, sealing, leakage and controls
Practical module limits	OPEN	Team-defined area, footprint, pressure drop, CAPEX, maintenance and lifetime gates
Durability and safety	OPEN	Wet cycling, creep, corrosion, fouling, electrical safety and long-duration mass balance
Immutable release	OPEN	Freeze reviewed source/output hashes in a clean commit or equivalent snapshot
Technology validation	NOT ACHIEVED	Measured coupled material, subsystem and full-system evidence

11 Recommended next experiments

CONVERT THE REMAINING UNCERTAINTY INTO MEASUREMENTS

11.1 Experiment 1: fixture qualification and blank closure

1. Assemble the 50×50-mm aperture cell with 2-mm gaps, metal load plates and double seals.
2. Map loaded gap and flatness against bolt torque; measure swept and dead volumes.
3. Run foil blanks over 3/6/12 L/min, 1/2/4-mm gaps and the intended pressure envelope.
4. Accept the fixture only when leakage, pressure-driven bypass and wall inventory fit within a predeclared fraction of the minimum useful polarity-odd signal.

11.2 Experiment 2: EO identification with matched zero-current control

1. Use a split common inlet and stabilise temperature, RH and pressure.
2. Run $+J/0/-J$ and mirrored ABBA blocks at multiple current levels.
3. Measure water on both sides, charge, terminal voltage and local temperatures; report odd, even and one-way terms.
4. Repeat with flow, gap, aperture area and perimeter changes.
5. Treat any current-off transport as the matched passive control, not as background to be assumed negligible.

11.3 Experiment 3: same-sample property bundle

For every candidate, freeze sample ID, batch, ion form, conditioning and wet thickness. Measure sorption/hysteresis, activity-resolved chemical mobility, EO drag, wet conductivity, compressed ASR, swelling/mechanics and the coupled two-sided flux. Candidate ranking begins only when all MAT-01–MAT-09 rows are complete for the same sample.

11.4 Experiment 4: receiver and thermal ledger

Choose one receiver concept and measure or bound water acceptance, pressure drop, pumping, regeneration/disposal, flooding/drainage, electrode compatibility and heat rejection. The pass criterion is not receiver SEC alone: the complete added ledger must remain inside the O30 conditional 0.13223-kWh/kg allowance with an uncertainty margin.

11.5 Experiment 5: measured-data topology replay

Only after Experiments 1–4, replay:

- post-ERV O30 on one common bus;
- higher completion with one bus;
- two voltage zones or staged EO if full-duty conflict remains;
- direct active EO in the ERV if measured passive transfer and films justify it.

The replay must include measured property covariance and uncertainty, not independent best-case axes.

Recommended sequence. Fixture qualification → matched passive/active coupon identification → same-sample property bundle → receiver/thermal ledger → measured-data topology replay. This sequence closes a decision gate at every step and avoids another unfocused model sweep.

12 Provenance and claim boundary

CANONICAL MACHINE-READABLE OUTPUTS, NOT COPIED REPORT ASSETS

This v8 report references the current figures and machine-readable evidence in place; no output figure or data file is copied into this report directory. The principal sources are:

- `outputs/project_closure_check_v1/closure_status.json`, `outputs/project_closure_check_v1/decision_gate_register.csv` and `outputs/project_closure_check_v1/source_manifest.json`;
- `outputs/final_delivery_check_v1/status.json`, `outputs/final_delivery_check_v1/gate_register.csv` and `outputs/final_delivery_check_v1/source_manifest.json`;
- `outputs/post_erv_full_eo_study_v1/`: `SUMMARY.md`, energy windows, transport rows and audit;
- `outputs/post_erv_eo_percentage_sweep_v1/`: percentage semantics, energy ledger, fixed-coordinate trajectory and audit;
- `outputs/workshop_common_voltage_diffusivity_v1/`: case register, refined envelope, bracket probe, profiles, audit and source manifest;
- `outputs/fom_assessment_v1/system_fom.csv`, `outputs/fom_assessment_v1/device_fom.csv`, `outputs/fom_assessment_v1/summary.json`, `outputs/fom_assessment_v1/audit.json` and `outputs/fom_assessment_v1/source_manifest.json`;
- `outputs/material_parameter_windows_with_nafion_v1/` and `outputs/direct_passive_nafion_erv_control_v1/`;
- `outputs/integrated_evidence_v1/material_data_requirements.csv` and `outputs/integrated_evidence_v1/next_study_matrix.csv`;
- `outputs/eo_vapor_cell_sizing_v1/audit.json`, `outputs/eo_vapor_cell_sizing_v1/source_manifest.json`, `outputs/eo_vapor_boundary_analysis_v1/audit.json` and `outputs/eo_vapor_boundary_analysis_v1/source_manifest.json`.

12.1 Permitted conclusions

- The model identifies conditional post-ERV and direct-ERV energy windows.
- O30 is a confirmed sampled common-voltage mechanism coordinate and the preferred first system-relevant coupon target.
- O50 needs no post-ERV EO duty at the declared absolute-humidity basis.
- Full post-ERV duty on one common bus below 2 V is not demonstrated in the sampled frozen coordinate and is not proven impossible.
- The material question is whether a coupled property bundle exists in one conditioned and durable sample.
- The model phase is sufficiently closed to begin material and subsystem feasibility work.

12.2 Prohibited conclusions

- No listed kW difference is a realised or validated saving.
- No FOM is an all-in product FOM unless every omitted term is added.
- No sampled material coordinate proves a continuous tolerance or a commercially available membrane.
- No current-density or area value is an acceptance limit in the present exploratory phase.
- Audit PASS is not material, module, system or commercial validation.

The report is a current decision synthesis, not an immutable release. Before external distribution, freeze the reviewed repository state, rerun the final delivery check and all declared tests, regenerate the PDF, record hashes and package the source with an explicit manifest.

A Compact numerical register

Metric	Value	Evidence class	Canonical source
AC-only equal-service power	1,166.400 kW	reference anchor	outputs/fom_assessment_v1/system_fom.csv
Passive ERV+AC power	543.740564 kW	modelled reference	outputs/fom_assessment_v1/system_fom.csv
Post-ERV O30 conditional subtotal	381.326719 kW	incomplete ledger	outputs/fom_assessment_v1/system_fom.csv
Post-ERV O30 conditional window	162.413845 kW	incomplete ledger	outputs/fom_assessment_v1/system_fom.csv
Post-ERV O30 known-core FOM	40.8641 kg/kWh	known core only	outputs/fom_assessment_v1/device_fom.csv
O30 common-voltage area at FOM point	7,173.55 m^2	model coordinate	outputs/fom_assessment_v1/device_fom.csv
O30 all-in break-even SEC	0.156701 kWh/kg	conditional budget	outputs/post_erv_full_eo_study_v1/break_even_by_target.csv
O30 remaining-duty completion	72.7003%	exact coordinate	outputs/post_erv_eo_percentage_sweep_v1/percentage_coordinates.csv
Historical 10% remaining-duty completion	22.1925%	traceability coordinate	outputs/post_erv_eo_percentage_sweep_v1/percentage_coordinates.csv
Full-duty conservative sink budget	0.128323 kWh/kg	conditional budget	outputs/project_closure_check_v1/closure_status.json
O30 common-voltage diffusivity span	10^{-14}–10^{-9} m^2/s sampled	one-factor screen	outputs/workshop_common_voltage_diffusivity_v1/diffusivity_envelope.csv
Full-duty refined sampled outcomes	21 physical / 28 unresolved	open topology gate	outputs/workshop_common_voltage_diffusivity_v1/diffusivity_envelope.csv
Active coupon aperture	50×50 mm	starting fixture	outputs/eo_vapor_cell_sizing_v1/fixture_dimension_basis.csv
Nominal coupon gap and flow	2 mm / 6 L min^{-1} $side^{-1}$	film mode	outputs/eo_vapor_cell_sizing_v1/channel_geometry_envelope.csv

End of report. Current interpretation as of 31 August 2026. All EO system comparisons remain conditional until measured material, receiver, thermal and hardware ledgers close.